

and SE12 (eq. (3.15b)). As verified in the numerical experiments in Section 7, the discretization of the nonlinear term proposed by [26] in SE12 has second-order accuracy, such that SE22 achieves second-order with a relatively simple approximation of the exponential functions; the application of (5.2) to the nonlinear term would lead to a much more complex formulation, since the integration factor acts both inside and outside the integral in (3.13).

In this section, we prove analytically the second-order convergence of the proposed discretization considering only the linear term. In order to provide a more complete comparative study with the original formulation of the SL-ETDRK schemes, as well as with SL-SI-SETTLS and the Eulerian ETDRK methods, we also present an estimate of the computing complexity and, in Section 6, a detailed linear stability analysis. In Section 7, we provide a numerical verification of these aspects considering both a one-dimensional advection equation and, with more details, the full nonlinear SWE on the rotating sphere.

It is worth noting that linear and interpolation operators do not commute in general (see Appendix of [26]), such that, even with $\varphi_0(\Delta t L/2)\varphi_0(\Delta t L/2) = \varphi_0(\Delta t L)$, we may have

$$\varphi_0(\Delta t L)[U^n]_*^n \neq \varphi_0\left(\frac{\Delta t L}{2}\right)\left[\varphi_0\left(\frac{\Delta t L}{2}\right)U^n\right]_*^n. \quad (5.8)$$

Finally, we highlight the similarities and differences of the proposed method w.r.t. the semi-Lagrangian Rosenbrock-type exponential schemes proposed by [35]. In both methods, a splitting of the time step integration is performed in order to achieve second-order accuracy. In [35], this is made by evolving the fluid particles along the Lagrangian trajectory from t_n to $t_{n+1/2} := t_n + \Delta t/2$, applying the exponential operator at $t_{n+1/2}$, and evolving the fluid particles from $t_{n+1/2}$ to t_n . Here, we consider a different splitting, with two applications of the exponential operator and a single application of the semi-Lagrangian approach; also, both applications of the exponential operator are identical, corresponding to a left multiplication by the constant matrix $\varphi_0(\Delta t L/2)$. Moreover, as described in Section 3.1, we consider a spectral discretization in space, which, with a proper rearrangement of the terms of the governing equations, allows for a straightforward computation of the matrix exponentials, while these are approximated in [35] using Krylov-subspace methods.

5.1. Truncation error analysis

The truncation error analysis of the proposed scheme is analogous to the first order scheme. We begin by considering the scalar case, *i.e.*, (5.3) applied to (4.1). It leads to the system of uncoupled ODEs

$$U_j^{n+1} = \varphi_0\left(\frac{\Delta t L_j}{2}\right)\left[\varphi_0\left(\frac{\Delta t L}{2}\right)U^n\right]_{*j}, \quad j = 0, \dots, P \quad (5.9)$$

We begin by inserting the exact solution u into the j -th equation of (5.9) and using the error estimate (4.5) for SETTLS using cubic Lagrange interpolation:

$$\begin{aligned} u_j^{n+1} &= \varphi_0\left(\frac{\Delta t L_j}{2}\right)\left[\varphi_0\left(\frac{\Delta t L}{2}\right)u^n\right]_{*j} \\ &= \varphi_0\left(\frac{\Delta t L_j}{2}\right)\varphi_0\left(\frac{\Delta t L_{e_j}}{2}\right)u_{e_j}^n + \mathcal{O}(\Delta t^2) + \mathcal{O}(\Delta t \Delta x^2) + \mathcal{O}(\Delta x^4) \end{aligned} \quad (5.10)$$

Then, using the exact solution (4.8) and the expression (5.2) for the integration factor along the Lagrangian trajectory Γ_j^{n+1} , we get the local truncation error

$$\begin{aligned}
\tau_j^{n+1} &= \frac{1}{\Delta t} \left[u_j^{n+1} - \varphi_0 \left(\frac{\Delta t L_j}{2} \right) \left(\frac{\Delta t L_{e_j}}{2} \right) u_{e_j}^n + \mathcal{O}(\Delta t^2) + \mathcal{O}(\Delta t \Delta x^2) + \mathcal{O}(\Delta x^4) \right] \\
&= \frac{1}{\Delta t} \left[e^{C_{2,j} \Delta t^3} e^{\frac{\Delta t L_{e_j}}{2}} e^{\frac{\Delta t L_j}{2}} u_{e_j}^n - e^{\frac{\Delta t L_j}{2}} e^{\frac{\Delta t L_{e_j}}{2}} u_{e_j}^n + \mathcal{O}(\Delta t^3) + \mathcal{O}(\Delta t \Delta x^2) + \mathcal{O}(\Delta x^4) \right] \\
&= \frac{1}{\Delta t} \left[(C_{2,j} \Delta t^3 + \mathcal{O}(\Delta t^6)) e^{\frac{\Delta t L_{e_j}}{2}} e^{\frac{\Delta t L_j}{2}} u_{e_j}^n + \mathcal{O}(\Delta t^3) + \mathcal{O}(\Delta t \Delta x^2) + \mathcal{O}(\Delta x^4) \right] \\
&= \mathcal{O}(\Delta t^2) + \mathcal{O}(\Delta x^2) + \mathcal{O} \left(\frac{\Delta x^4}{\Delta t} \right)
\end{aligned} \tag{5.11}$$

confirming that the proposed method is second-order accurate.

In the vectorial case (Eq. (5.5) applied to (5.4)), this result is also true if $\mathbf{L}_j$ and $\mathbf{L}_{e_j}$ commute. Indeed, in this case, the last equality in (5.2) would hold and $\varphi_0(\Delta t \mathbf{L}_j/2)$ and $\varphi_0(\Delta t \mathbf{L}_{e_j}/2)$ would commute, such that the third equality in the truncation error analysis (5.11) would be true. In the general case of a non-commutative linear operator $\mathbf{L}$ along each Lagrangian trajectory, a more careful analysis would be required. For non-commuting square matrices A, B of same size, it holds

$$e^{A+B} - e^A e^B = \frac{1}{2} [BA - AB] + \text{third-order terms} \tag{5.12}$$

such that, with $A = -\Delta t \mathbf{L}_j/2$ and $B = -\Delta t \mathbf{L}_{e_j}/2$, the proposed approximation (5.2) to the integration factor would have a leading error term of order $\mathcal{O}(\Delta t^2)$, which would contribute with $\mathcal{O}(\Delta t)$ to the local truncation error, thus ensuring only first-order accuracy. Note, however, that if $\mathbf{L}$ is smooth enough, we can expand $\mathbf{L}_j$ around $\mathbf{L}_{e_j}$ along Γ_j^{n+1} to get

$$\begin{aligned}
\mathbf{L}_{e_j} \mathbf{L}_j - \mathbf{L}_j \mathbf{L}_{e_j} &= \mathbf{L}_{e_j} \left(\mathbf{L}_{e_j} + \Delta t \frac{d\mathbf{L}}{dt} \Big|_{e_j} + \mathcal{O}(\Delta t^2) \right) - \left(\mathbf{L}_{e_j} + \Delta t \frac{d\mathbf{L}}{dt} \Big|_{e_j} + \mathcal{O}(\Delta t^2) \right) \mathbf{L}_{e_j} \\
&= \Delta t \left(\mathbf{L}_{e_j} \frac{d\mathbf{L}}{dt} \Big|_{e_j} - \frac{d\mathbf{L}}{dt} \Big|_{e_j} \mathbf{L}_{e_j} \right) + \mathcal{O}(\Delta t^2) = \mathcal{O}(\Delta t)
\end{aligned} \tag{5.13}$$

such that

$$e^{-\frac{\Delta t}{2} [\mathbf{L}_j + \mathbf{L}_{e_j}]} - e^{-\frac{\Delta t}{2} \mathbf{L}_j} e^{-\frac{\Delta t}{2} \mathbf{L}_{e_j}} = \frac{\Delta t^2}{8} [\mathbf{L}_{e_j} \mathbf{L}_j - \mathbf{L}_j \mathbf{L}_{e_j}] + \mathcal{O}(\Delta t^3) = \mathcal{O}(\Delta t^3) \tag{5.14}$$

This implies that, if the integration factor can be written in the matrix exponential form (5.2), then it actually satisfies

$$P_{n_j}(t_{n+1}) = e^{-\int_{t_n}^{t_{n+1}} \mathbf{L}_{x_j(s)} ds} = e^{-\frac{\Delta t}{2} [\mathbf{L}_j + \mathbf{L}_{e_j}] - C_{2,j} \Delta t^3} = \left(e^{-\frac{\Delta t \mathbf{L}_j}{2}} e^{-\frac{\Delta t \mathbf{L}_{e_j}}{2}} + C_j \Delta t^3 \right) e^{-C_{2,j} \Delta t^3} \tag{5.15}$$

for a given constant C_j accounting for the non-commutation of $\mathbf{L}_j$ and $\mathbf{L}_{e_j}$. In other words, the splitting of the exponential would introduce an $\mathcal{O}(\Delta t^3)$ term that would contribute with $\mathcal{O}(\Delta t^2)$ to the local truncation error, thus preserving the second-order accuracy. Nevertheless, it is important

to notice that, in the case where $\mathbf{L}_{\mathbf{x}(t)}$ and $\mathbf{L}_{\mathbf{x}(s)}$ do not commute, we would have to additionally assume that the integration factor exists and can be written in the matrix exponential form (3.14), otherwise, the development of the local truncation error (5.11) would not be valid, since we would not be able to write an explicit form to the exact solution u_j^{n+1} in (4.8). However, the numerical simulations presented in Section 7 indicate that second-order accuracy is achieved by (5.5) even if this commutation is not asserted. Finally, note that commutation-related problems are known issues in semi-Lagrangian schemes (*e.g.*, the non-commutation between Laplace transforms and linear differential operators along Lagrangian trajectories, as studied in [22]).

5.2. Computational complexity

We now provide an estimate of the computational complexity of the proposed modified semi-Lagrangian exponential methods and the other time integration schemes considered here. In each time step, the following computations are performed:

- In the case of exponential methods, the evaluation of the φ_k and/or the ψ_k functions. Since these functions are defined recursively, their cost increases for larger k . These functions need to be computed for each pair of wavenumber (m, n) , and we recall that the matrix exponentials are computed in this work considering the diagonalization of the matrix (eq. (3.8)), such that we need only to compute the exponential of the eigenvalues of $\mathbf{L} = \mathbf{L}_{\mathbf{G}}$ (two per wavenumber pair);
- In the case of semi-Lagrangian schemes, the trajectories need to be estimated, and spatial interpolations to the departure points need to be performed;
- In the case of SL-SI-SETTLS (3.11), the semi-implicit discretization for the linear term requires both the application of $\mathbf{L}$ and $\mathbf{L}^{-1}$, the latter through the solution of a Helmholtz equation [40];
- Finally, all schemes require the evaluation of at least one nonlinear term. We take into account reutilizations of the computed terms between consecutive time steps.

Table 1 summarizes the number of each operation listed above performed by each time integration scheme. In this table, each φ_k/ψ_k function is depicted individually; $\mathbf{x}_d$ and $()_*$ denote respectively the evaluation of the Lagrangian trajectories and the interpolation to the departure points; $\mathbf{L}$ and $\mathbf{L}^{-1}$ correspond to the semi-implicit scheme described above; and $\mathbf{N}_A$ and $\mathbf{N}_R$ indicate the evaluation of the (respectively advection and remaining) nonlinear terms.

Scheme	φ_0	φ_1	φ_2	ψ_1	ψ_2	$\mathbf{x}_d$	$()_*$	$\mathbf{L}$	$\mathbf{L}^{-1}$	$\mathbf{N}_A$	$\mathbf{N}_R$
ETD1RK	1	1	0	0	0	0	0	0	0	1	1
ETD2RK	1	1	1	0	0	0	0	0	0	2	2
SE11	1	0	0	1	0	1	1	0	0	0	1
SE12	2	0	0	1	2	1	2	0	0	0	2
SE21	3	0	0	1	0	1	2	0	0	0	1
SE22	4	0	0	1	2	1	3	0	0	0	2
SL-SI-SETTLS	0	0	0	0	0	1	2	1	1	0	1

TABLE 1. Number of operations performed per time step by each time integration scheme. $\mathbf{x}_d$ and $()_*$ denote respectively the evaluation of the Lagrangian trajectories and the interpolation to the departure points; $\mathbf{L}$ and $\mathbf{L}^{-1}$ correspond to the semi-implicit scheme.